Phase shift between current and voltage across a capacitor

The phase shift φ between current and voltage across a capacitor at the cutoff frequency ω_{cutoff} is _____

The phase shifts for some other frequencies are:

Part II – RLC parallel circuit

1. Parallel RLC resonant circuit

$C =$ 100 nF $R =$ 1 kΩ $L =$ 10 mH $U_{\text{in}} =$ 1V / 2.8 V_{pp}

From these values we get: $\omega_0^{\text{theory}} =$ 31'623 $\frac{\text{rad}}{\text{s}}$

f kHz	
φ ($\omega = 2\pi f$)	3, 3.15, 3.42, 3.59, 3.81, 4.02, 4.57, 4.76, 4.81, 5.18, 5.4, 5.5, 5.7, 6.04, 6.28, 6.45, 6.74, 7.6, 5.32, 5.26, 5.35, 5.06, 5.17])
U_{out} V_{max}	([0.34, 0.38, 0.44, 0.5, 0.58, 0.66, 0.96, 1.08, 1.1, 1.28, 1.3, 1.28, 1.2, 1.04, 0.91, 0.86, 0.74, 0.54, 1.3, 1.3, 1.3, 1.24, 1.28])
φ deg	[72.6, 71, 67, 65, 62, 58, 40, 32.4, 30, 8, -5.5, -11.5, -22.6, -38, -45, -49.2, -55, -65, -1.3, 2.3, -2.5, 16.6, 9.5])

2. Graphical representation of U_{out} as function of ω

3. Quality factor Q of the resonant circuit

(determine either using a fit and/or by graphical representation

$Q_{\text{graphical}} =$ 2.99 $\pm$ _____ $Q_{\text{theory}} \approx 3.16$

$Q_{\text{fit}} =$ 3.00 $\pm$ _____ *(only D-PHYS)*

4. Current through a resonant circuit

$C =$ 100 nF $R =$ 1 kΩ $L =$ 10 mH $U_{\text{in}} =$ 1V / 2.8 V_{pp}

f kHz	
ω $k\frac{\text{rad}}{\text{s}}$	[[4.23 4.6 5.02 5.08 5.17 5.27 5.4 5.58 5.84 6.14]
	[26.57787385 28.90265241 31.54159024 31.91858136 32.48406804 33.11238657 33.92920066 35.06017401 36.69380219 38.57875779]
I_R A	[3.01 3.79 4.76 4.85 4.97 5.02 5. 4.84 4.43 3.82]
	[-85. -74. -50. -43. -34. -4. 48. 65. 75. 85.]]
φ deg	

Graphical representation of φ as function of ω.

5. (only D-PHYS) Resonance frequencies for various L and C

i) $C =$ 10 nF ; $L =$ 1 mH ; $\omega_0 =$ 307 k $\frac{rad}{s}$

ii) $C =$ 100 nF ; $L =$ 1 mH ; $\omega_0 =$ 99 k $\frac{rad}{s}$

iii) $C =$ 1 µF ; $L =$ 1 mH ; $\omega_0 =$ 31 k $\frac{rad}{s}$

iv) $C =$ 10 nF ; $L =$ 10 mH ; $\omega_0 =$ 103 k $\frac{rad}{s}$

v) $C =$ 100 nF ; $L =$ 10 mH ; $\omega_0 =$ 33 k $\frac{rad}{s}$

vi) $C =$ 1 µF ; $L =$ 10 mH ; $\omega_0 =$ 10 k $\frac{rad}{s}$

Graphical representation of ω_0 as *meaningful* function $f(L, C)$.

6. (only D-PHYS) Square and triangular signal

Supply the RLC circuit with a rectangular or triangular signal and change U_{in}. What do you observe?

keywords: Ring down, Fourier transform of the input signal

at resonance we see sinus-shaped waves for both inputs. at 10x resonance frequency as well, but way smaller. The fourier-transform has peaks at odd multiples of resonance ω_0 so $\omega_0, 3\omega_0, 5\omega_0 \ldots$ they get smaller. because of band pass we only see one of them => sinus shape

7. (voluntary) Currents in a parallel circuit

Measure the currents I_L and I_C with the multimeter. What do you observe? How large are the currents relative to the current across the resistor I_R??